Lab 3: Component Modelling & Architectural Pattern Selection

Self-Service Coffee Kiosk System

SRN: PES1UG24AM439

Section 1 — UML Component Diagram

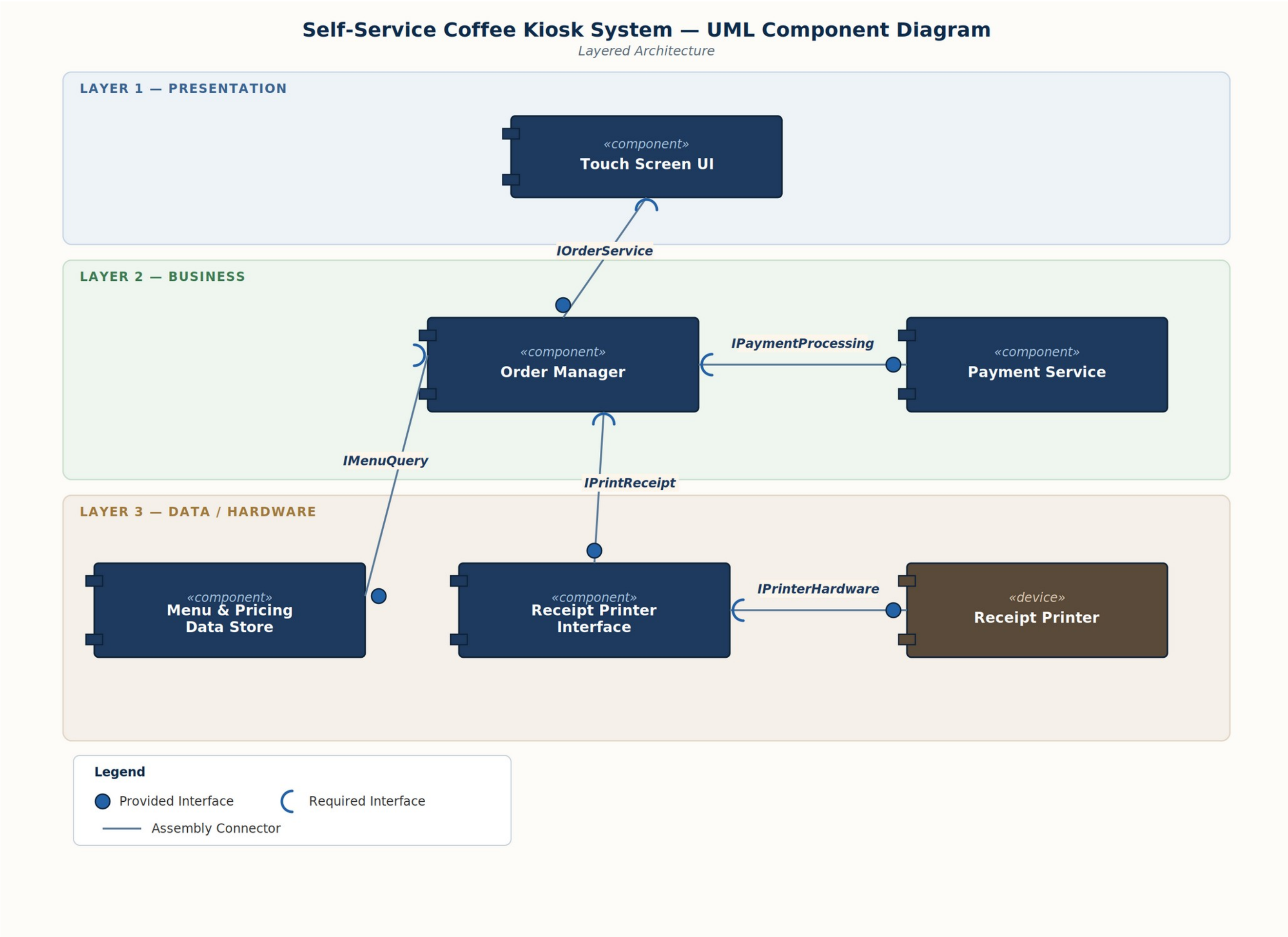

Figure 1 — Layered component diagram showing 5 software components and 1 external hardware device, connected through 5 provided/required interfaces.

Section 2 — Written Justification

1. Architecture Selection

We chose Layered Architecture for the Self-Service Coffee Kiosk System.

2. Architectural Choice

The system is organised into three layers, each communicating only with the layer directly above or below it through well-defined interfaces. The Presentation layer contains the Touch Screen UI. The Business layer contains the Order Manager and the Payment Service, which together handle order creation and payment logic. The Data/Hardware layer contains the Menu & Pricing Data Store, the Receipt Printer Interface, and the physical Receipt Printer device. This separation keeps user interaction, business logic, and data/hardware access independent of one another.

3. Reason 1 — Matches a Single-Device, Fixed-Scope System

The kiosk is a standalone device that serves one customer at a time with a fixed, narrow set of responsibilities — menu selection, payment, and receipt printing. A microservices architecture would introduce network latency and operational overhead such as service discovery and independent deployment pipelines, which a single-device kiosk with constrained hardware does not need. Similarly, a Client-Server architecture is unnecessary here because there is no multi-client access pattern for the kiosk to support; all processing happens locally on one device.

4. Reason 2 — Clean Mapping to Natural Responsibilities

The kiosk's requirements divide naturally into presentation (touchscreen interaction), business logic (order processing and payment handling), and data/hardware access (menu and pricing storage, and receipt printing). A Layered Architecture keeps each of these concerns cleanly separated behind explicit component interfaces, which makes the system easier to build, test, and modify — for example, swapping the receipt printer model only affects the Receipt Printer Interface component, without rippling through the rest of the system.

5. Security Advantage

All payment-card handling is confined to the Payment Service component in the Business layer. The Touch Screen UI in the Presentation layer never accesses raw card data directly — it only exchanges order and payment-result information with the Payment Service through the Order Manager's IPaymentProcessing interface. This isolates sensitive payment processing behind a single, well-defined boundary and reduces the surface area for tampering or interception at the UI layer.

6. Performance Benefit

The Order Manager in the Business layer can retrieve and hold menu and pricing information from the Menu & Pricing Data Store through the IMenuQuery interface for the duration of an order. Repeated touchscreen selections during the ordering flow can then be served from this in-memory data rather than issuing a fresh data-store query for every screen interaction, keeping the UI responsive on the kiosk's constrained hardware. Because the architecture is a simple layered structure rather than a distributed set of services, it also avoids the network overhead that inter-service calls would otherwise add.